

Raft under Physical Communication Models: What Relativity Changes, and What It Does Not

Tomasz Kazulak
Independent study

Structured literature survey · 2026-09-25

Abstract. Consensus protocols such as Raft are specified against abstract communication models, while physical systems add constraints those models do not name: a finite signal speed, clocks that measure proper time, and no observer-independent simultaneity. We survey 60 studies, identified by targeted searches and forward citation chasing, to establish what is known about Raft under non-ideal and physically grounded communication models. Three findings emerge. First, Raft’s safety depends on no timing assumption and has recently been shown to survive relativistic causality. Second, the large literature on Raft under loss, partitions and variable delay concerns liveness and performance, and relativity adds nothing to it beyond a lower bound and a time dependence on delay. Third, the one Raft mechanism whose safety depends on clocks, the leader lease, has been analysed only under bounded clock drift, an assumption that is not frame-invariant. No study examines leases, or clock-based coordination generally, in relativistic spacetime. We identify this as the open problem and state candidate questions for an analytical follow-up.

1 Introduction

Distributed-systems courses present consensus against an idealised network in which messages are delayed arbitrarily but eventually delivered, and processes share no clock [1], [2]. A physicist reading these models naturally asks whether they survive contact with the physical world. Signals propagate no faster than light. Clocks on moving or gravitationally separated nodes accumulate different proper times [3], [4]. And relativity denies a global notion of “now”, on which specifications such as linearizability [5] appear to depend. Lamport’s logical clocks were themselves motivated by special relativity [6], yet whether a concrete, widely deployed protocol stays correct under physical constraints has only recently been studied formally [7], [8], [9].

We take Raft [10] as the object of study. It is widely deployed and precisely specified, and it separates cleanly into a timing-independent safety core and timing-dependent mechanisms. The survey’s aim is to fix the boundaries of current knowledge before any new analysis is attempted: what has been established, what follows directly from established models, and what is open.

The contributions are (i) a map of the communication models under which Raft and equivalent protocols have been analysed; (ii) a classification of Raft’s timing-dependent mechanisms by whether they affect safety or only liveness; (iii) a synthesis of the small literature on distributed computing in relativistic spacetime; and (iv) the identification of clock-based coordination under relativistic time as the open problem.

2 Scope and research questions

The object is Raft and leader-based crash-fault-tolerant state-machine replication whose results transfer to Raft. We consider communication models ordered by the constraints they impose:

- synchronous;
- partially synchronous [2];
- asynchronous [1];
- asynchronous with loss, omission and partial partitions;
- delay- and disruption-tolerant networks [11];
- relativistic spacetime, in which delivery respects the light-cone structure and each process measures its own proper time.

Out of scope are time-transfer engineering without a computational model, quantum communication, and blockchain systems that use Raft as a black box.

RQ1 Under which communication models have Raft and equivalent protocols been analysed, for which properties, and with which methods?

RQ2 Which Raft mechanisms depend on timing or clock assumptions, for safety or only for liveness, and how are those assumptions stated?

RQ3 Which models of distributed computation in relativistic spacetime exist, and what correctness results do they establish?

RQ4 To what extent have clock-dependent mechanisms been analysed under relativistic or otherwise physically grounded models of time?

3 Method

This is a structured survey rather than a full systematic review. It follows the reporting logic of PRISMA 2020 [12] without claiming its complete procedure, and it uses snowballing as described by Wohlin [13].

Identification. Sixteen targeted web-search queries produced 149 records. We removed non-scholarly pages and duplicates, screened 73 records by title and abstract, and included 34. Backward citation chasing from these and from the Raft literature added 19 foundational studies.

Forward snowballing. Eight seeds were chosen as the studies most central to the question that also have few enough citations to screen completely. Five come from the relativistic and timekeeping strand [4], [7], [8], [9], [14], and three concern Raft under network faults and leases [15], [16], [17]. Litmaps returned 56 citing records. They collapse to 39 distinct works once preprint and published versions are merged. Of these, 5 were already included, 7 were newly included, and 27 were excluded as off-topic: optics, AI for systems research, cluster operations, and essays on computation and physics. A ninth intended seed, Mattern [18], is not indexed by Litmaps.

Result. The final corpus has 60 studies. Title and abstract screening was performed by an AI assistant and accepted by the author (Section 4). Table 1 summarises the flow.

Table 1: Study selection.

Stage	Records
Targeted searches: records identified	149
After removing non-scholarly records and duplicates	73
Included from searches	34
Added by backward citation chasing	19
Forward snowballing: citing records (8 seeds)	56
distinct works after merging versions	39
newly included	7
Final corpus	60

4 Use of AI assistance

This survey was conducted with an AI assistant, Claude (Anthropic), used through a chat interface connected to web search, the author’s Zotero library and arXiv. Claude Code was used to set up the software environment. We report the division of work because it bears directly on how far the findings can be trusted.

Performed by the assistant:

- proposing the research questions, search strings and eligibility criteria;
- running the targeted searches;
- screening titles and abstracts, including the forward-snowballing results;
- resolving bibliographic metadata and adding records to Zotero;
- drafting the text of this article;
- the preliminary lease argument in Section 6 and a numerical check of it.

Performed by the author:

- posing the motivating question and deciding the scope;
- choosing to reduce a full systematic review to a structured survey;
- confirming the snowballing seeds and running the forward chase in Litmaps, a citation index rather than a generative tool;
- approving each stage, and taking responsibility for the result.

Verification status. Bibliographic records were resolved from DOIs or arXiv identifiers wherever these existed; records without them were flagged. Statements about individual studies were checked against titles, abstracts and search excerpts, not against full texts. A few characterisations, of results that are standard in the field, rest on the assistant’s background knowledge. These are the timing requirements and read mechanisms of Raft, and the network semantics of its machine-checked proof. Full-text verification of every claim attributed to a cited study is pending.

The principal risks are misattributed or overstated claims, screening errors that a second human reviewer would catch, and a uniform framing inherited from a single drafting voice.

5 Results

5.1 RQ1: Communication models and methods

Table 2 maps the corpus. The division of labour is sharp. Safety is analysed by proof under the asynchronous model, and it has been machine-checked for Raft under a network that drops, duplicates and reorders messages [19]. Everything else, from lossy links to wireless and space deployments, is analysed for liveness, latency or availability, by simulation, analytical modelling or testbeds.

Table 2: Communication models under which Raft and equivalent protocols have been analysed.

Model	What is analysed	Studies
Asynchronous	Safety of the time-free core; machine-checked proofs	[10], [19]
Partially synchronous	Liveness of leader election; timing requirement that broadcast time be much less than the election timeout	[2], [20]
Omission, partial connectivity	Livelock and leaderlessness under partial partitions; PreVote and CheckQuorum counterexamples; real-world partition failures	[15], [16], [21], [22]
Unreliable channels	Which channel failures consensus can tolerate; tight bounds via generalised quorum systems	[23], [24]
Probabilistic loss and delay	Split probability, commit latency and reliability as functions of loss rate and timeouts	[25], [26], [27], [28], [29]
Variable wide-area delay	Adaptive or learned election timeouts; randomised protocols whose liveness does not rest on timeouts	[30], [31], [32], [33]
Wireless, edge, space	Raft variants and evaluations for vehicular, edge and satellite links; DTN architecture	[11], [34], [35], [36], [37]
Relativistic spacetime	Correctness conditions and safety proofs under causal (light-cone) delivery	[7], [8], [9]

5.2 RQ2: Timing-dependent mechanisms

Raft uses time in four places (Table 3). Election timeouts, heartbeats, and the PreVote and Check-Quorum extensions affect only availability: a badly tuned timeout causes needless or failed elections but never two leaders in one term [10], [20]. The exception is the leader lease used to serve linearizable reads without a quorum round-trip. Its safety requires that the leader’s lease expire, on the leader’s clock, before any follower’s election timeout could have let a new leader emerge. That in turn needs a bound on how far the two clocks’ rates can diverge [20].

The lease literature states this assumption as a bounded drift rate relative to real time. This holds for the original lease mechanism [38], for the general treatment of clocks in distributed protocols [39], and for recent Raft-specific designs, including a TLA+-verified Raft lease [17] and a production design

for many consensus groups [40]. Adversarial manipulation of clock rates is treated as a threat to lease safety [41]. ReadIndex, the quorum-based alternative, uses no clock and is safe asynchronously [20].

Table 3: Where Raft depends on time.

Mechanism	Affects	Timing assumption
Election timeout, heartbeat	Liveness	Partial synchrony; timeout well above round-trip time
PreVote, CheckQuorum	Liveness	As above; known liveness gaps under partial connectivity
Leader lease (lease reads)	Safety	Bounded clock-rate divergence between leader and followers
ReadIndex reads	Neither	None (one quorum round-trip)

5.3 RQ3: Distributed computing in relativistic spacetime

The strand is small, continuous and recent. Mattern observed that vector time in distributed systems has the structure of Minkowski causality [18]. Matherat and Jaekel showed that synchronous and asynchronous circuits mirror Newtonian and relativistic causal structures [14], [42]. Gilbert and Golab were the first to ask what linearizability means when operations are only partially ordered by causality, and proposed relativistic variants of it [7].

Jayanti gave a single execution model covering classical and relativistic systems, and showed that correctness established for asynchronous algorithms carries over [8]. Aeini, Golab, Olivetti and Ruppert proved the strongest of Gilbert and Golab’s conditions for Raft [9], and Jayanti and Natarajan extended the causal model to quantum systems [43]. Forward chasing of all seeds surfaced no further work in this strand.

A separate engineering literature quantifies relativistic timekeeping: its magnitude in navigation and communications [3], [4], [44], and its effect on information freshness [45], [46]. None of it contains a model of distributed coordination.

5.4 RQ4: Clock-based coordination under physical time

No study in the corpus analyses a clock-dependent consensus mechanism under a relativistic model of time. The relativistic results cover asynchronous algorithms, whose correctness makes no reference to clocks. Aeini et al. explicitly name algorithms that use physical clocks for coordination as future work [9]. The lease studies, conversely, assume bounded drift relative to a common real time [17], [40], and the timekeeping studies have no protocol model. The two literatures do not meet.

6 Discussion

We separate what is established, what follows directly from established models, and what is open.

Established.

- Raft’s safety is independent of timing and holds under relativistic causality [9], [19].
- Its liveness requires partial synchrony [2] and fails in specific, well-characterised ways under partial connectivity [15], [16].
- Its only timing-dependent safety mechanism is the leader lease [20].

Directly derivable. Three consequences follow from the models above without new results.

- A network that delivers messages only into the sender’s future light cone produces a subset of asynchronous executions, so every asynchronous safety argument transfers. This is the intuition behind [8].
- Physics bounds message delay from below, by distance over c , and makes it time-varying. This instantiates partial synchrony rather than replacing it, so for liveness relativity is “just delay”.
- Table 4 compares the sizes involved. The largest motion-induced effect is first order in $\beta = v/c$ and is kinematic: a changing propagation delay. Genuinely relativistic rate effects are second order, or of order $\Delta\Phi/c^2$ for gravity, and lie orders of magnitude below the tolerance of commodity oscillators.

Table 4: Orders of magnitude of rate effects relevant to timeouts and leases.

Effect	Order	Typical size
Changing propagation delay (first-order Doppler)	β	5×10^{-5} between LEO satellites; 10^{-4} at Earth’s orbital speed
Time dilation	$\beta^2/2$	$\approx 3 \times 10^{-10}$ in low Earth orbit
Gravitational rate difference	$\Delta\Phi/c^2$	$\approx 4.5 \times 10^{-10}$, GPS orbit vs. ground [3]
Commodity quartz oscillator tolerance	—	10^{-5} to 10^{-4}

Open. What does not reduce to delay is the relativity of simultaneity, and it bites exactly where a protocol converts elapsed time on one clock into a claim about another node. “Bounded drift between two clocks” presupposes a simultaneity convention for comparing distant clocks. It is therefore not a frame-invariant assumption, and the lease safety argument has no frame-invariant statement in the literature.

A preliminary observation, to be verified rather than relied upon, suggests what such a statement might look like. The causal chain that could make a lease read stale passes through the follower’s timer. By the reverse triangle inequality of Minkowski spacetime, an inertial leader’s proper time along that chain is at least the follower’s timeout. If so, first-order Doppler effects cancel, and only a leader whose worldline is not a geodesic, whether accelerating or held static in a gravitational field, can lose lease time: a twin-paradox effect.

7 Threats to validity

The survey had a single human reviewer, and screening and drafting were AI-assisted (Section 4), with full-text verification still pending. Identification relied on targeted web search rather than on Scopus or Web of Science, and the queries were partly seeded with known items, which biases toward known work. Forward snowballing depends on Litmaps’ index: one intended seed was not indexed, and the most recent seed has no citations yet. The relativistic strand is small enough that we believe the forward chase covers it. The Raft-under-faults strand is large, and it is represented rather than exhausted.

8 Conclusion and next steps

Physics changes little about Raft. Its safety core survives relativistic causality, and its liveness sees physics only as delay. The single point of contact is the leader lease, the one mechanism that turns local elapsed time into a claim about other nodes. Its safety assumption has never been stated in frame-invariant form. We take the following candidate questions to the analytical phase:

1. What is the frame-invariant condition under which a Raft leader lease is safe?
2. In which physically realistic configurations, such as satellites, ground–orbit links and interplanetary distances, does that condition differ measurably from the classical bounded-drift rule?
3. Does a lease-based read satisfy the strongest relativistic linearizability condition that Raft satisfies without leases [9]?

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